OK

OMNIKON

NATIONAL HACKATHON 2026

TEAM NAME

HarvestGuard

MEMBERS

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THEME

Software / AgriTech

PROBLEM STATEMENT

Reducing Grain Spoilage in Rural Storage — Poor storage conditions in rural warehouses lead to significant grain spoilage every year. Create an app that helps monitor and optimize storage conditions to minimize losses.

ONE MISSION. BUILD THE IMPOSSIBLE.

Problem Understanding

10–20%

of the global grain harvest is lost every year to improper storage

Why grain spoils in rural storage

- Moisture above 13–14% creates ideal conditions for mold and fungal growth (aflatoxin risk)
- Temperature & humidity fluctuation accelerates biological breakdown of grain
- Pest and rodent infestation goes undetected until visible, physical damage
- Rural warehouses rely on manual, reactive inspection — no early warning system
- Most tech fixes assume sensor budgets rural storage units simply don't have

Proposed Solution

GrainGuard AI — Predictive Spoilage Risk Advisor

- Zero-hardware MVP: a photo of the grain heap + basic conditions is enough to compute a live Spoilage Risk Score
- AI vision catches early mold, discoloration, or pest damage — before it's visible to the naked eye
- Combines grain-science thresholds with hyperlocal weather forecasts to warn before a risk spike hits, not after
- Delivers one clear instruction, in the worker's own language, with voice readout — not a data dashboard they have to interpret

Key Features

PIC

Photo Risk Scan

Snap the grain heap, get an instant risk score — no setup needed

AI

Early Defect Detection

Vision model flags mold, discoloration & pest damage early

WX

Weather-Aware Forecast

Hyperlocal forecast data predicts risk before it spikes

VOX

Voice + Local Language

Clear spoken instructions for low-literacy workers

IoT

Optional Sensor Mode

Plug in ESP32 + DHT22 for live continuous monitoring

DASH

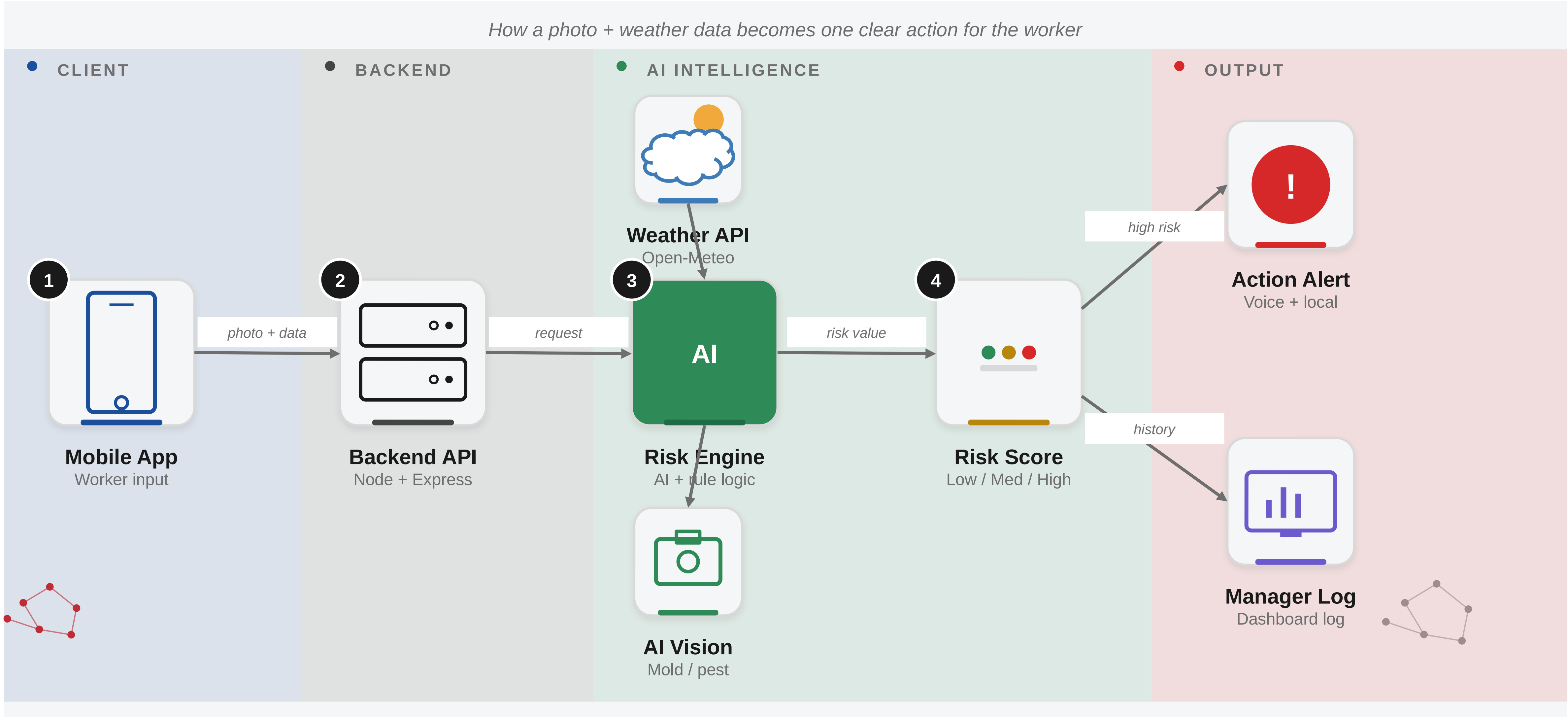
Manager Dashboard

Track risk history across multiple storage units over time

Technology Stack

Frontend	React Native / Flutter — cross-platform, runs on low-cost Android phones
Backend	Node.js + Express — API, auth, orchestration
AI / ML	MobileNet image model + rule-based Spoilage Risk Engine
Data & APIs	Firebase / MongoDB + Open-Meteo weather API (free tier)
Hardware (optional)	ESP32 + DHT22 sensor tier for live continuous readings

System Architecture



Implementation Plan

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Phase 1

Hackathon MVP

Photo-based risk scoring, core app, local-language alerts

Phase 2

Pilot Deployment

Test with 1–2 real warehouses, refine AI on real photos

Phase 3

Sensor Integration

Add ESP32 + DHT22 live data tier for equipped warehouses

Phase 4

Scale & Expand

Roll out to PACS / FCI network, add more regional languages

Feasibility & Impact

Feasibility

- No new hardware needed for MVP — pilot-ready on existing smartphones
- Risk logic based on established agri-science, not experimental methods
- Free-tier APIs keep pilot running cost near zero

Challenges & Strategies

- Low connectivity → offline-first mode, syncs when back online
- Low tech comfort → large icons, voice-guided steps
- Manager trust → advisory-first design, human stays in control

Potential Impact

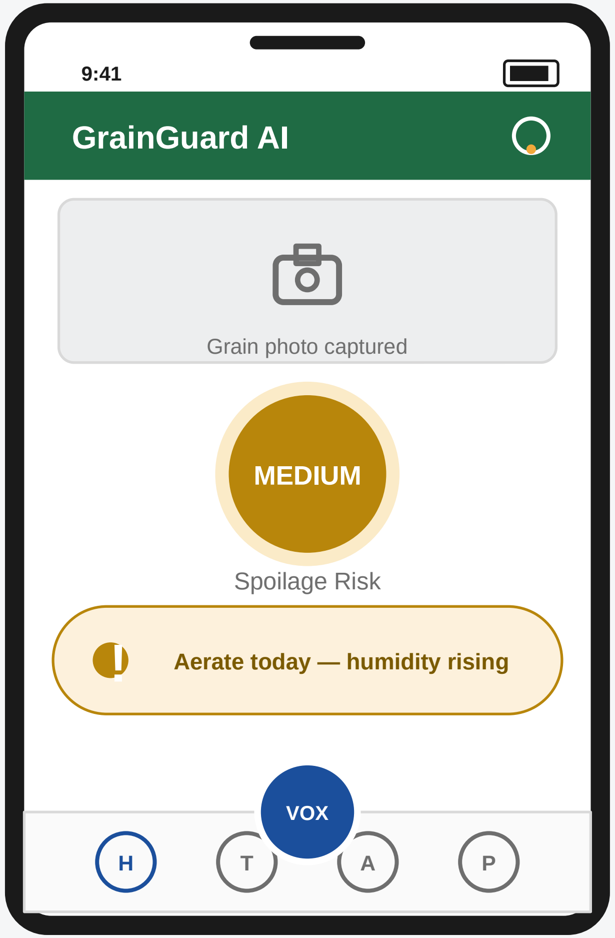
- Helps small farmers, warehouse workers, PACS/FCI-style managers
- Catches spoilage risk days before visible damage — still savable
- Removes the "can't afford sensors" barrier most tech ignores

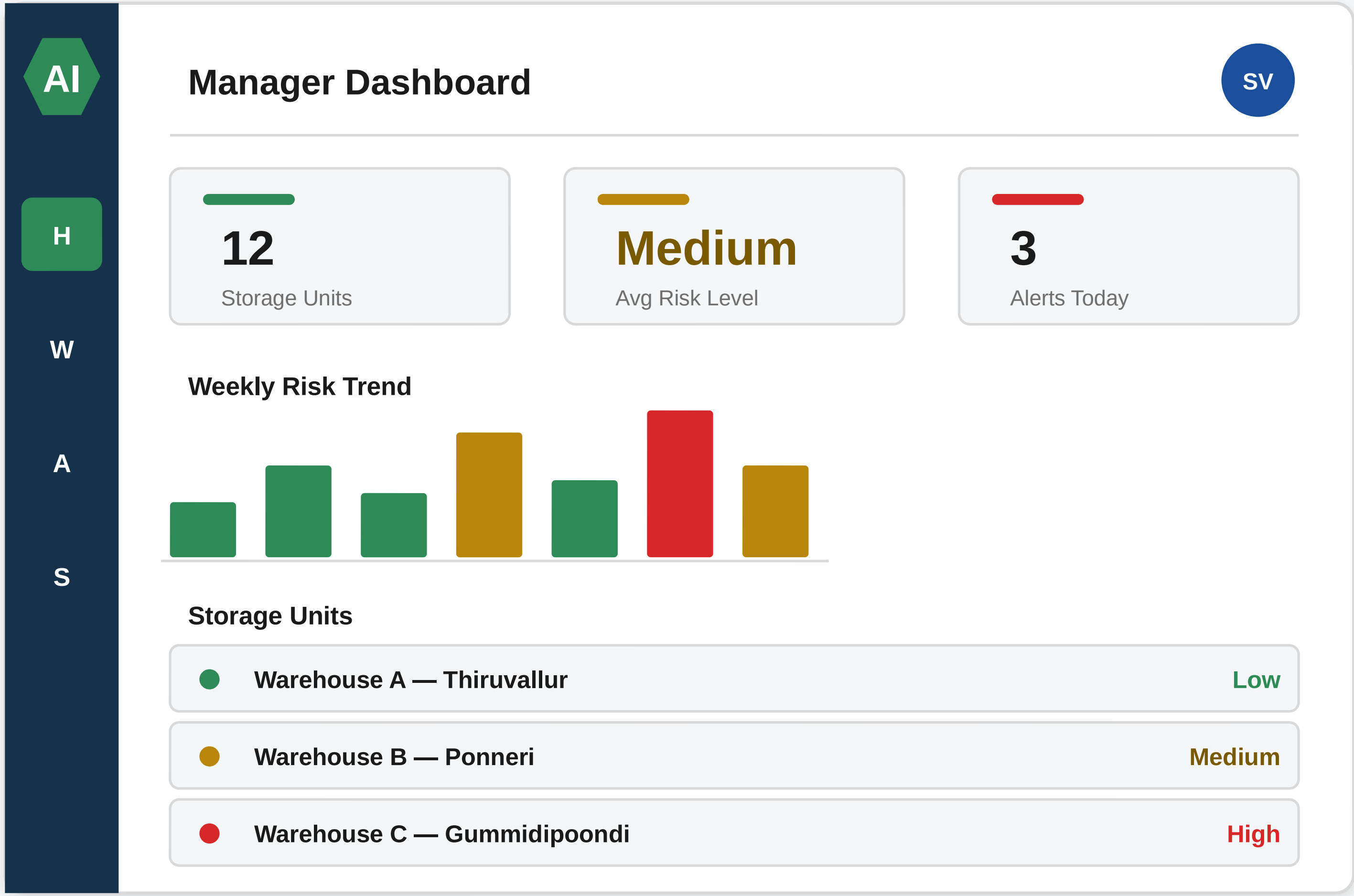
Benefits

- Economic: cuts into the 10–20% of harvest lost to poor storage
- Social: local-language, voice-guided, genuinely usable
- Scalable: village shed today, full FCI network tomorrow

Prototype Preview

Early UI concept — worker app on the left, manager dashboard on the right





Research and References

- Post-harvest grain storage loss data — global loss of 10–20% of harvest annually due to improper storage/handling (industry & agri-research sources)
- Grain moisture & temperature thresholds for spoilage prevention (safe moisture below 13–14%, aeration/fumigation practices) — SlideShare: "Grain Spoilage Minimization at Storage"
- Aflatoxin / fungal risk in stored cereals — post-harvest spoilage management literature
- Weather data source: Open-Meteo API (free, open-access hyperlocal forecasts)
- Comparison baseline: IoT-based cold-chain / grain monitoring patterns (ESP32 + DHT22 + SMS alerts) — used to justify our no-hardware-first approach